

Jahangirnagar University



Assignment Topic

Project Evaluation and Feasibility Study: Jahangirnagar University
Student Information System (JUSIS).

Course Title: Software Project Management and Quality Assurance
Course Code: PMSCS 660

SUBMITTED TO

Dr. Mohammad Zahidur Rahman
Professor, Department of Computer Science and
Engineering

SUBMITTED BY

Md. Habibur Rahman
Id No: CSE202601013
Department of Computer Science and Engineering

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Project Evaluation Report: Jahangirnagar University Student Information System (JUSIS)

Part-A

Programme Context

1. Programme vs project justification:

JUSIS is not just a single project; it is a strategic programme. It consists of a group of projects that are managed in a coordinate way to gain benefits that would not be possible were the projects to be managed independently. Two subprojects under this programme are:

Admission module: For online application and enrolment.

Examination module: For digital grading and transcript generation.

2. Programme mandate:

This is initial planning document is the programme mandate. For JUSIS, the

mandate is not to deliver a new capability to centrally manage all student records, replacing manual files. This perfectly fits JU's organizational goals of becoming a "Digital University"

3. Vision statement & Blueprint:

Vision statement: JUSIS will provide the new capability to seamlessly manage academic and administrative data in real-time.

Blueprint: TO obtain this capability the changes to be made include installing new central servers, upgrading campus networks, and retraining administrative staff

Part - B

Benefits management plan

Providing an organization with a capability does not guarantee that this will provide

benefits envisaged. There are 6 expected benefits for JUSIS

Category

Example benefits for JUSIS

- | | |
|---|---|
| A) Quantified and valued | Automating transcript generation
3 admin staffs 10 W/W.
$\text{value} = 3 \times 10 \times \text{BDT } 500/h \times 52 W$
$= \text{BDT } 780000/y$ |
| A) Quantified and valued | Paperless forms reduce printing costs.
$\text{value} = 15000 \text{ appx} \times \text{BDT } 50$
$= \text{BDT } 750000/y$ |
| B) Quantified but not valued | A 50% decrease in manual data entry errors across all dept. |
| B) Quantified but not valued | A 60% reduction in student wait times for registration inquiries |
| C) Classified but not easily quantified | Improved public reputation for the university due to modern system. |
| C) Classified but not easily quantified | A more motivated workforce as staff are freed from repetitive manual filing. |

Benefit monitoring: Benefits management has to be outside the project because the project will have been completed. The programme director will be responsible to monitor achievement of benefits post-launch.

Part - C

Cost-Benefit Analysis & cash flow Forecasting

1. Cash Flow Table:

year	Cost (BDT)	Benefits BDT	Cash Flow (BDT)	Accumulated Net (BDT)
0	4500000	0	-4500000	-4500000
1	600000	800000	200000	-4300000
2	600000	1200000	600000	-3700000
3	600000	1800000	1200000	-2500000
4	600000	2200000	1600000	-900000
5	600000	2600000	2000000	1100000

2. Total net profit: Net profit is the value of all cash-flows for the lifetime of the application

$$\begin{aligned}\text{Total profit} &= (-4.5M + 200K + 600K + 1.2M \\ &\quad + 2.0M) \\ &= \text{BDT } 1100000\end{aligned}$$

3. Payback period: This is the time it takes to start generating a surplus of income over outgoings.

→ At the end of year 4, the deficit is BDT 900000

→ Year 5 cash flow is BDT 2000000.

→ Remaining time = $(900000 / 2000000) = 0.45y$

→ $0.45y \times 12m = 5.4 \text{ months}$.

→ Result: 4 years and 5 months.

4. Return on Investment (ROI) Formula:
 $(\text{Average annual profit} / \text{Total investment}) \times 100$;

→ Average annual profit = $1100000 / 5 = \text{BDT } 220000$

→ $\text{ROI} = (220000 / 4500000) \times 100 = 4.88\%$

Part - D

Net present value (NPV) & Internal rate return

i. Calculate NPV: Discount factor $\$1/(1+r)^{t_1}$

where $r = 0.12$

year	Cash-flow (BDT)	Discount factor	Discount cash flow (BDT)
0	-4500000	1.0000	-4,500,000
1	200000	0.8929	178571
2	600000	0.7972	478316
3	1200000	0.7118	854136
4	1600000	0.6355	1,016,830
5	2000000	0.5674	1,134,854

Total

NPV = -837,293

Calculation:

Conclusion:

- The project is not financially worthwhile under a 12% discount rate over 5 years. The negative NPV means the returns are lower than what the university could earn by simply investing the money elsewhere at 12% interest.

2. Interpret IRR:

Internal rate of return is the discount rate that would produce an NPV of 0 for the project.

If the IRR is 18% and borrowing rate is 12%, the university should proceed. This means the project generates an 18% return easily covering the 12% cost of borrowing the funds, ultimately creating financial value.

Part - E

Risk Evaluation & Decision Analysis

1. Project Risk Matrix:

We can draw up a project risk matrix to assess risks.

<u>Risk</u>	<u>Likelihood</u>	<u>Impact</u>
1. Resistance from administrative staff	High(H)	High(H)
2. Legacy data migration errors	Medium(M)	High(H)
3. Development budget exceeded	Medium(M)	Medium(M)

Migration strategy: A project may fail not through poor management but because it should never have been started. To avoid this, we will ensure end-user and heavily involved from the beginning and provide thorough training programs. So they accept the new system rather than resist it.

2. Decision Tree Analysis:

Assuming a standardized financial benefit of BDT 10,000,000 for a successful system.

→ Build (local developers): cost = 3M, Success = 80%.

$$EMV = (0.8 \times 10,000,000) - 3,000,000 = \text{BDT } 5,000,000$$

→ Buy (offer-the-self): cost = 5M, Success = 95%

$$EMV = (0.95 \times 10,000,000) - 5,000,000 = \text{BDT } 4,500,000$$

Recommendation: The university should choose to build. Although buying off-the-shelf is safer, building in-house has a higher expected monetary value (EMV). Projects with the highest potential returns are often most risky.